

New aboveground biomass equations by components for small black spruce in peatland ecosystems of Western Canada

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Abstract

Black spruce is the most common tree species in Canada, dominating treed peatlands where they are usually stunted. We used 495 destructively sampled trees from 56 plots to develop allometric models of aboveground biomass by components (stem, branches, and needles) for small (<5 m tall) black spruce from peatlands in the Taiga Plains and Boreal Plains Ecozones of Western Canada, for which there were no specific models available of biomass by components. We used leave-one-plot-out cross-validation to assess transferability and compare our models with existing national and ecozone-specific equations. Our models predicted total tree biomass with better accuracy and less biased estimates than the national model (relative RMSE: 30% versus 35% national; relative bias: +1% versus -12% national). Similar results were obtained in other external datasets. Existing ecozone equations performed worse than either our models or the national ones. When we applied the models at the plot level to predict aboveground biomass density ($\text{Mg}\cdot\text{ha}^{-1}$), our models outperformed the national model again (relative RMSE: 15.9% versus 18.6% national, relative bias: +3.5% versus -13.6% national). These results indicate that at least for peatlands of Western Canada, our models provide better aboveground biomass estimates for small black spruce trees than existing models.

Key words: allometry, biomass modelling, black spruce, peatlands, boreal forest

1. Introduction

Accurate tree biomass measurements are essential for estimating the amount of carbon stored by forests. These carbon stock estimates can then be used to adjust sustainable forest practices to enhance the forest carbon sink, monitor ecosystem-level changes, and provide valuable inputs for climate models (Chave et al. 2014). Directly measuring forest biomass density ($\text{Mg}\cdot\text{ha}^{-1}$) requires harvesting, drying, and weighing all trees within an area, which is impractical (Brown 1997). Instead, allometric equations are used to estimate the aboveground biomass (AGB, the dry weight of live trees excluding their roots) of individual trees using physical characteristics like stem diameter at a height of 1.3 m (diameter at breast height, DBH) and tree height as predictor variables (Brown 1997). These equations are typically species-specific and calibrated using destructively sampled trees from the geographic domain where the equation will be applied, which can range from local (hundreds of km^2 or smaller) to regional (tens of thousands of km^2) to national (hundreds of thousands to millions of km^2). Owing to their sensitivity to environmental factors (White et al. 2016), these equations can lead to large estimation errors when applied over a large

area (Calders et al. 2015). Competition for resources amongst neighbouring trees can also affect tree growth. Alterations in crown dimensions, height, and DBH have been observed based on the number of trees within a plot (del Río et al. 2019). In addition to stand density, climate and age have also been shown to influence tree growth (Forrester et al. 2017b), indicating that variables external to the individual tree could improve biomass predictions.

In Canada, the most-used allometric AGB equations were put forth by Lambert et al. (2005a) and later updated by Ung et al. (2008). They were fitted using a dataset consisting of nearly 9000 trees from 33 species sampled from plots across Canada as part of the Energy from the Forest Biomass Conversion (ENFOR) Program (Ung et al. 2017). While these models have shown to be relatively accurate (relative root mean square error [RMSE] <30% of the mean tree AGB), Canada's National Forest Inventory (NFI) uses a customized version of these equations for their ground plots that was fitted using a subset of the ENFOR dataset that consists of only trees from the ecozone where the ground plot is located (Lambert et al. 2005b). These ecozone versions are assumed to better reflect allometric relations for trees in that particular environment,

as corroborated (with some exceptions) by Case and Hall (2008). However, a customized version fitted with a small sample of trees may be less reliable than the national version, which was fitted with considerably more trees. In addition, both the customized and the national models could be less accurate in ecosystems where tree morphology may differ. One such example is treed peatlands, which are common across the Canadian boreal forest and cover 13% of the country's land surface (Tarnocai et al. 2011). These areas are characterized by poor soil drainage resulting in water saturation and anaerobic conditions leading to peat accumulation (Tarnocai and Stolbovoy 2006). Peatlands are often dominated by black spruce (*Picea mariana* (Miller.) Britton, Sterns & Poggenburgh) (Liefers 1986; Wieder 2006), and because of their acidic and waterlogged terrain, tree growth is often stunted in comparison with upland forests (Smith et al. 2007; Beaulne et al. 2021). As such, we hypothesize a peatland-specific model for small (between 1.3 and 5 m tall; NB. 1.3 m is the minimum height required to measure DBH, and 5 m is the threshold customarily used to distinguish trees from shrubs) black spruce trees will outperform both the national and ecozone-customized models that include black spruce of all sizes and from mostly upland ecosystems, since the ENFOR program was geared towards bioenergy from harvested forests. Furthermore, these models could be enhanced with the inclusion of plot-level variables, a possibility worth exploring.

The development of a peatland-specific biomass model for black spruce is particularly important given that black spruce is by far the most common leading species in Canada, occupying over 200 Mha or >50% of Canada's treed area (Hermosilla et al. 2022), and is commonly found in peatland environments (OECD 2010). In this study, we used a new dataset consisting of 495 black spruce trees <7.7 m tall, destructively sampled from within 56 temporary plots in peatlands from Western Canada. The DBH, height, and dry weight of the stem, needles, and branches of each of the harvested trees were measured in the laboratory. Our specific objectives were to (1) develop individual tree models of AGB by components (stem, branches, and needles) specific to small peatland black spruce for the Taiga Plains (TP) and Boreal Plains (BP) Ecozones, separately by ecozone and combined; (2) assess the transferability of the developed models through leave-one-plot-out (LOPO) cross-validation and through application to external tree datasets; (3) test whether the AGB estimates from our models are significantly better than those from the national (Ung et al. 2008) and, where appropriate, the NFI customized (Lambert et al. 2005b) equations as well as other existing models; (4) upscale the estimates from our combined model and the national model to plot-level AGB in $\text{Mg}\cdot\text{ha}^{-1}$ and compare their respective accuracies; and (5) incorporate stand variables (such as basal area and stem density) and assess the impact they have on biomass predictions at the individual tree and plot levels.

2. Materials and methods

2.1. Study area

The TP Ecozone is mostly located in the Northwest Territories (NWT) and consists of over half a million km^2 of low-lying

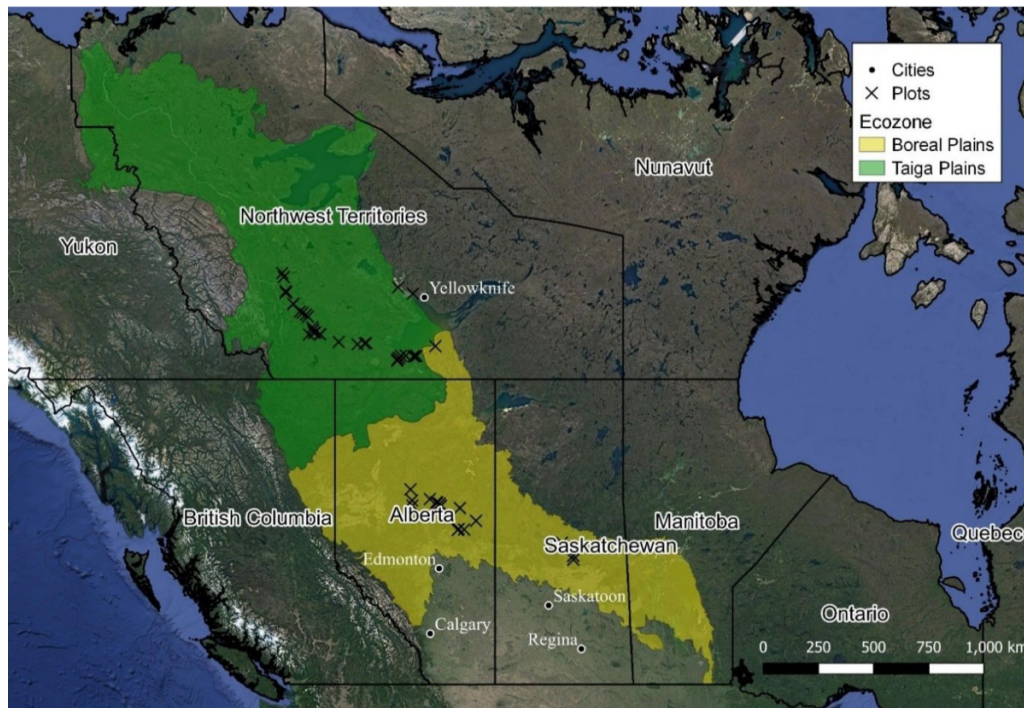
plains along Canada's largest river, the Mackenzie (Deh'cho in Dene, Kuukpak in Inuvialuktun), half of which are peatlands and one-fifth of which is water (Ecosystem Classification Group 2009). Summers are cool and short, and snow persists more than half a year. Precipitation is low to moderate, averaging 250 to 500 mm per year (Ecosystem Classification Group 2009). Annual temperatures in the TP Ecozone depend highly on latitude, and average between 6.5 °C and 14 °C in the summer and between -26 °C and -15 °C in the winter (Ecological Stratification Working Group 1995). The BP Ecozone starts east of the Canadian Rockies, south of the TP, and stretches roughly 0.7 million km^2 , straddling British Columbia, Alberta, Saskatchewan, and Manitoba (Natural Resources Canada 2021). The region averages temperatures between -2 °C to 2 °C annually, with mean summer temperatures ranging between 13 °C and 15.5 °C and winter temperatures between -17.5 °C and -11 °C. It averages between 300 and 625 mm of precipitation annually, and peatlands make up between 25% and 50% of the BP Ecozone (Ecological Stratification Working Group 1995).

2.2. Field data collection

A total of 35 plots were established in the NWT from 2019 to 2021, and another 23 (20 in Alberta and 3 in Saskatchewan) were established in 2004 and 2016 (Fig. 1) and used to develop the models (objective 1). All plots from the TP were circular and 50 m^2 (3.99 m radius) in size except for two that were 25 m^2 (2.82 m radius) because of higher than usual stem density. The BP plots were established under a different project and were either 7.07 by 7.07 m square plots (50 m^2 , 6 plots established in 2016) or 4 by 4 m square plots (16 m^2 , 17 plots established in 2004). A total of 495 harvested trees were used in this study and they ranged from 1.3 to 7.7 m in height (interquartile range [IQR], or difference between upper and lower quartiles, 3.1 – 1.7 = 1.4 m), 0.1 to 6.9 cm in DBH (IQR 3.1 – 1.3 = 1.8 cm), and 0.11 to 14.75 kg in AGB (IQR 2.05 – 0.65 = 1.4 kg). Per plot summary statistics for the harvested trees can be found in the Supplementary Materials (Table S1), as well as a scatterplot showing the heights and DBHs of the trees used in this study (Fig. S1). The plots in NWT were located along the highway system in the southern portion of the territory, with none being established further north than a latitude of 63.2°N owing to the lack of an all-season road. Candidate stands were visually selected from the highway while travelling; as such, all plots were established in stands that were visible from the vehicle, but at least 20 m away from the edge of the road corridor. The stand selection process aimed to sample unproductive black spruce peatlands with an average height of <5 m, while also representing the breadth of heights, densities, and ages therein. Pure black spruce stands were preferred, but some incidental larch (*Larix laricina* (Du Roi) K. Koch) was acceptable.

The species, status (dead or alive), height, DBH, and crown base were measured for every tree ≥ 1.3 m in height. In addition, the number of trees <1.3 m tall and their heights were recorded by species in a "small tree" subplot of 0.56 m radius (not used in this study). Two of the 35 plots in NWT contained

Fig. 1. Map of Western Canada overlaid with the Taiga Plains (green) and Boreal Plains (yellow) Ecozones. Black X's denote the locations of the plots used in this study. Ecozone shapefile: A national Ecological Framework for Canada (Agriculture and Agri-Food Canada 2013). Provinces/territories boundaries shapefile: Statistics Canada (2016). Basemap: imagery © 2023 TerraMetrics, map data © 2023 Google.



only trees shorter than 1.3 m, hence they were excluded. Ten trees per plot were selected, harvested, and bagged for biomass separation and weighing in the laboratory. The selection tried to mimic the height distribution of the plot. For that purpose, trees were grouped into height classes, with the first class being for trees 1.30–2.49 m tall, then increasing in 1 m intervals and the last being for trees ≥ 5.5 m. Then at least one tree per class present in the plot was selected, adding more trees for more frequent classes. Care was taken to ensure the trees selected for destructive sampling adequately represented the various heights and tree forms of the site. If fewer than 10 trees taller than 1.3 m were present in the plot, they were all destructively sampled (this happened in only one of the plots).

A Haglof Vertex IV hypsometer was used to measure height to the nearest decimetre. DBH measurements were recorded to the nearest millimetre using a digital micrometer or a Lufkin Executive diameter tape. The harvested trees were also marked at 1.3 m to indicate the DBH location for laboratory measurements. The trees were then cut into pieces to fit into 1 or more 12" x 38" polypropylene sacks for transport and storage.

Green weight was measured for each harvested tree sack using a Salter 1400 digital scale with a precision of 1 g (for sacks less than 3 kg) or a Yamato Accu-Weigh T-50 tubular scale with a precision of 250 g (for sacks greater than 3 kg). The weight of the empty sacks (54 g) was not included in these measurements.

2.3. Laboratory measurements

Once the harvested tree sacks were returned to the laboratory, they were put in a storage room either in Hay River, NWT, or in Edmonton, AB and left to dry for at least a week until there was <1% difference in weight over a 24 h period. The temperature of the storage room was between 60 °C and 65 °C. Once the trees were dried, the stem was reconstructed to measure length (laboratory-measured height) and weight. It was also measured at the DBH mark to obtain laboratory-measured DBH. Large branches were initially separated by hand, and then a series of makeshift sieves of various sizes were used to separate the medium-sized debris. A combination of colanders, strainers, and baskets were used during this phase. The finer material necessitated manual separation by spreading it across a table and extracting the components by hand (needles, small twigs, and sloughed bark; the latter was added to the stem weight). Each component (needles, branches, and stem) was then weighed to the closest 100th of a gram, which provided reference data for model development.

2.4. Model development

Three AGB models were created for objective 1, each corresponding to a different geographic domain: (1) the TP model, based on 33 plots from the TP; (2) the BP model, based on 23 plots from the Boreal Plains; and (3) the Western Peatlands (WP) model, based on all 56 plots both in the TP and

the BP (Table S2). Each model comprises three AGB equations for stem, branches, and needles, respectively. Total AGB is derived as the sum of those three components. Modelling followed Lambert et al. (2005a) and was undertaken using the MODEL procedure in SAS version 3.8 (SAS Institute Inc. 2020) following a seemingly unrelated regression (SUR) model (Zellner 1962). SUR fits a separate equation for each component using correlated error terms and adds the constraint that the sum of some components must minimize the estimation error of the sum of those components. In our case, crown (needle + branches), no branches (needle + stem), and total (crown + stem) were used as constraints. The general form of the equation was as follows:

$$(1) \quad AGB_i = \beta_i \cdot DBH^{\alpha_i} \cdot H^{\omega_i} + \varphi_i$$

where AGB_i (kg) is the biomass of component i , DBH (cm) and H (m) are the predictor variables, and β , α , ω , and φ are the coefficients to be fitted by the SUR process.

During preliminary tests, we found that height was not a significant predictor (p values for the height coefficient >0.05) in the equations for needle and branch AGB in all three models (TP, BP, and WP), leading to equations of the form

$$(2) \quad AGB_i = \beta_i \cdot DBH^{\alpha_i} + \varphi_i$$

for these two components, where β , α , and φ are model coefficients as in eq. 1.

Initial results indicated that the residuals grew in magnitude with the size of the tree. We accounted for this heteroscedasticity by giving less weight to the residuals of large trees than to those of smaller trees following the method in Lambert et al. (2005a). This method assumes that the residuals derived from ordinary least squares (OLS) are proportional to a power of tree DBH . The corresponding equation for the residuals of each biomass component was fitted as

$$(3) \quad \ln e^2 = b + c \cdot \ln DBH$$

where e is the residual from the unweighted model, and b and c are the intercept and slope of the log-transformed linear regression, respectively. Weighted residuals are then calculated as

$$(4) \quad \frac{Actual - Predicted}{\sqrt{DBH^c}}$$

where c is calculated in the regression shown in eq. 3. Finally, the SUR coefficients are obtained by minimizing the sum of squares of the weighted residuals subject to the aforementioned SUR constraints. The effect of using weights in the SUR model can be appreciated with a comparison of AGB predictions with laboratory-measured AGB for both weighted and unweighted models (Figure S2).

Tree measurements were made in the laboratory with precise instruments, and measurement uncertainty for the laboratory measurements for each of the three variables used in this study was observed to be less than 0.2% of its average

measurement ($\delta\text{height} = 0.02\%$, $\delta\text{DBH} = 0.02\%$, $\delta\text{AGB} < 0.01\%$). As such, measurement uncertainty was considered to be negligible. Model uncertainty is reported in the results section as the standard error for each coefficient, as well as regression fit statistics (RMSE, bias, mean absolute error [MAE], and adjusted R^2).

Once models were created using all the trees from all the plots in the TP, BP, and WP datasets, we assessed their transferability (objective 2) using LOPO cross-validation (Wenger and Olden 2012), wherein the model is fitted with all plots but one, and then used to estimate the AGB of the trees in the left-out plot. This process is repeated until all the trees in the dataset have an estimate, which is then compared with the laboratory-measured AGB (by component and in total). This method of cross-validation is suited to assess model transferability to new locations not used in the modelling (Wenger and Olden 2012). As a second test of transferability, the TP and BP models were applied to each other's ecozone and compared in terms of RMSE and bias. Finally, to assess model robustness (i.e., how similar the model parameters or predictions are across LOPO models), RMSE, bias (mean signed residual of trees in the left-out plot), MAE, and adjusted R^2 (when regressing observed (y) versus predicted (x) AGB values) were computed from the estimates of each LOPO iteration. The standard deviation of the set of RMSE values in total tree AGB from the LOPO iterations was computed for each of our three models (TP, BP, and WP). LOPO coefficients were compared to the overall coefficients that were fitted using all plots to calculate the percentage of LOPO coefficients that were less than 5% different than the overall coefficients.

Multiple comparisons were made between the models developed in this study and the existing national (Ung et al. 2008) and, where applicable, customized NFI models (objective 3). RMSE, bias, MAE, and the adjusted R^2 of the observed versus predicted values were compared between models. To better understand how these models perform as tree size increases, absolute and signed residuals of the WP model were also compared with those of the national model for each quartile of laboratory-measured AGB. Further comparisons between the WP model and other existing AGB models for black spruce (Barney et al. 1978; Yarie et al. 2007; Fradette et al. 2021; Wagers et al. 2021) based on different tree datasets can be found in the Appendix.

2.5. Upscaling to plot level

As part of objective 4, the estimates for total (sum of components) biomass from both the WP model and the national model were upscaled to calculate plot-level AGB (a.k.a. AGB density, in $\text{Mg} \cdot \text{ha}^{-1}$). Because only 10 trees in a given plot were harvested and dry weighed, we computed the equivalent area taken up by the harvested trees as follows:

$$(5) \quad A'_i = A_i \cdot (N_H/N_i)$$

where A'_i is the equivalent area of the plot when only the harvested trees are considered, A_i is the true area of the plot, N_H is the number of harvested trees taken from the plot, and N_i is the total number of trees in the plot. The sum of the laboratory-measured AGB of the harvested trees from plot i

is then divided by A_i' (eq. 5) to provide a reference value of AGB density in that plot, which was converted to $\text{Mg}\cdot\text{ha}^{-1}$. While this reference value may not be the best estimate of the true AGB density in the plot, it is a good surrogate for this upscaling exercise given the harvesting selection process explained in Section 2.2. Four of the plots that were established in 2004 did not have data on stem density or total number of trees and were excluded from this analysis. We then compared plot AGB density estimates with the plot reference values to assess bias and RMSE at the plot level for both the WP and national models.

2.6. Effect of stand variables on individual tree level and plot-level estimation

Due to the potential impact that ecological factors like competition can have on tree allometry, and to pursue objective 5, we fitted a new set of models that included stand attributes as predictors. Out of the 56 plots, 17 (from the 2004 study) lacked data to calculate stem density and/or basal area and were excluded from this part of the study; the remaining 39 plots contained 372 harvested trees. Stem density was calculated by dividing the total number of trees taller than 1.3 m within a plot by the plot size and then scaled to $\text{trees}\cdot\text{ha}^{-1}$. Basal area (amount of area within the plots occupied by tree trunks) was calculated as the sum of the cross-section at breast height (1.3 m) of the trees in the plot and then scaled to $\text{m}^2\cdot\text{ha}^{-1}$. Models were fitted following the methods outlined in Section 2.4 and these combinations of predictors:

$$(6) \quad \text{AGB}_i = \beta_i \cdot \text{DBH}^{\alpha_i} \cdot H^{\omega_i} \cdot \text{SD}^{\zeta_i} + \varphi_i$$

$$(7) \quad \text{AGB}_i = \beta_i \cdot \text{DBH}^{\alpha_i} \cdot H^{\omega_i} \cdot \text{BA}^{\psi_i} + \varphi_i$$

$$(8) \quad \text{AGB}_i = \beta_i \cdot \text{DBH}^{\alpha_i} \cdot H^{\omega_i} \cdot \text{SD}^{\zeta_i} \cdot \text{BA}^{\psi_i} + \varphi_i$$

where AGB_i is the biomass of component i , DBH (cm), H (m), SD ($\text{stems}\cdot\text{ha}^{-1}$), and BA ($\text{m}^2\cdot\text{ha}^{-1}$) are the predictor variables, and β , α , ω , ζ , ψ , and φ are the coefficients fitted by the SUR process. To allow for a comparison on the same dataset, eq. 1 was also fitted with just these 372 trees.

Estimates of AGB were obtained for each model and tree using the LOPO cross-validation technique outlined in Section 2.4. RMSE, bias, and adjusted R^2 of observed versus predicted values were derived for each model and compared. Additionally, the Akaike information criterion (AIC) and Bayesian information criterion (BIC) were calculated for each of the models to assess whether adding in the extra predictors improved the model fit.

To assess the effect that stand attributes related to stem density and mean tree size have on plot-level errors, we linearly regressed both signed and absolute plot-level AGB errors with stem density, stand height, mean quadratic DBH, and basal area. A significant (at $\alpha = 0.05$) slope coefficient would indicate there is a significant relationship between that ecological variable and the magnitude and/or direction of plot-level errors.

3. Results

3.1. Objective 1: develop peatland-specific AGB models

Three models, each consisting of three equations (needles, branches, and stem), were created in this study: one for the TP Ecozone, one for the BP Ecozone, and one that encompassed both the TP and BP together (WP model). Table 1 shows the resulting coefficients for the three models.

3.2. Objective 2: assess model transferability

Model transferability was assessed using LOPO cross-validation. All LOPO models showed relatively little variation among them, with the standard deviation of RMSE values across the 56 LOPO models being smaller than 20% of the mean tree AGB for all datasets ($\sigma_{\text{WP, RMSEs}} = 0.27 \text{ kg}$ (15%), $\sigma_{\text{TP, RMSEs}} = 0.33 \text{ kg}$ (18%), $\sigma_{\text{BP, RMSEs}} = 0.22 \text{ kg}$ (or 12% of mean tree AGB)). Furthermore, differences between the LOPO coefficients and the final model coefficients of the WP model show that 95% or more of the LOPO values for most coefficients are $\leq 5\%$ different from the final values (Supplementary Materials, Figure S3). The three coefficients where this is not the case (ω_{stems} , β_{branches} , β_{needles}) still showed differences of $\leq 5\%$ in 93%, 86%, and 88% of the LOPO values, respectively. This indicates that the models remained consistent throughout the cross-validation and that even if another peatland plot were added to the dataset, the final coefficients are likely to be $\leq 5\%$ different than the values given here.

When tested in each other's ecozones, the TP and BP models performed similarly, with relative RMSE differences of less than 3% (in TP: $\text{rRMSE}_{\text{TP}} = 32\%$, $\text{rRMSE}_{\text{BP}} = 34\%$; in Beal Plains: $\text{rRMSE}_{\text{TP}} = 26\%$, $\text{rRMSE}_{\text{BP}} = 29\%$) and relative bias magnitude differences of less than 10% (in TP: $|\text{rBias}_{\text{TP}}| = 1\%$, $|\text{rBias}_{\text{BP}}| = 11\%$; in BP: $|\text{rBias}_{\text{TP}}| = 6\%$, $|\text{rBias}_{\text{BP}}| = 10\%$) when compared to the model built specifically for that ecozone.

We saw further evidence of the transferability of our model, when the WP model was applied to an Alaskan dataset (Mack 2014) (Section A2 in the Appendix), where it outperformed the models presented in Yarie et al. (2007) and Barney et al. (1978) in terms of RMSE. It also outperformed the Barney model in terms of bias. These models use basal diameter as sole predictor as opposed to DBH and height, so this could explain in part why our model performs better. The Yarie model was fitted using a dataset that included the trees used for the comparison as well. These results suggest that the WP model can also be used in Alaskan peatlands. The results of applying the WP model to black spruce trees from an open woodland dataset (Section A4 in Appendix) indicate it may not be fully transferable to a very different ecosystem, although it performed slightly better than the national model in that dataset. This and the fact that the WP model performed similarly to the national model in the subset of small black spruce trees included in the dataset originally used to fit the national model (Section A3 in Appendix) suggest the WP model may be preferable to the national model on any small black spruce trees regardless of ecosystem.

Table 1. Model coefficients for the Taiga Plains Ecozone, Boreal Plains Ecozone, and Western Peatlands region developed in this study.

Coefficient	Taiga Plains	Boreal Plains	Western Peatlands
β_{stem}	0.0704 ± 0.0039	0.0780 ± 0.0080	0.0556 ± 0.0032
α_{stem}	1.9293 ± 0.0442	1.6473 ± 0.0949	2.0049 ± 0.0515
ω_{stem}	0.5727 ± 0.0372	0.6088 ± 0.1013	0.6151 ± 0.0464
Φ_{stem}	0.1892 ± 0.0159	0.1255 ± 0.0274	0.2079 ± 0.0169
β_{branches}	0.0145 ± 0.0027	0.0365 ± 0.0039	0.0325 ± 0.0035
α_{branches}	2.5615 ± 0.1075	2.3438 ± 0.0653	2.1427 ± 0.0640
Φ_{branches}	0.1103 ± 0.0056	0.1320 ± 0.0056	0.1094 ± 0.0040
β_{needles}	0.0107 ± 0.0029	0.0447 ± 0.0057	0.0341 ± 0.0048
α_{needles}	2.4283 ± 0.1538	2.1308 ± 0.0808	1.8696 ± 0.0868
Φ_{needles}	0.1171 ± 0.0054	0.1238 ± 0.0808	0.1136 ± 0.0043

Note: *The models follow the form $AGB = \beta \cdot DBH^{\alpha} + \Phi$ for needles and branches and $AGB = \beta \cdot DBH^{\alpha} \cdot H^{\omega} + \Phi$ for stems (AGB, aboveground biomass; DBH, diameter at breast height; H, height). Each coefficient is presented alongside its standard error (SE) as calculated during the model-fitting process.

Table 2. Accuracy metrics* for the aboveground biomass (AGB) models developed in this study† and the national‡ and NFI§ models (RMSE, root mean square error; MAE, mean absolute error).

Model	RMSE (kg)	Bias (kg)	MAE (kg)	Adjusted R ²
<i>Dataset: Western Peatlands (495 trees)</i>				
Western Peatlands	0.53 (30%)	0.02 (1%)	0.34 (19%)	0.93
National	0.60 (33%)	−0.22 (−12%)	0.42 (23%)	0.92
<i>Dataset: Taiga Plains (311 trees)</i>				
Taiga Plains	0.57 (32%)	−0.01 (0%)	0.34 (19%)	0.93
Western Peatlands	0.57 (32%)	0.07 (4%)	0.35 (20%)	0.93
National	0.63 (35%)	−0.19 (−11%)	0.41 (23%)	0.92
NFI Taiga Plains	1.02 (57%)	−0.70 (−39%)	0.70 (39%)	0.93
<i>Dataset: Boreal Plains (184 trees)</i>				
Boreal Plains	0.53 (29%)	0.10 (5%)	0.35 (19%)	0.91
Western Peatlands	0.47 (25%)	−0.06 (−3%)	0.32 (17%)	0.92
National	0.55 (30%)	−0.27 (−15%)	0.43 (24%)	0.91
NFI Boreal Plains	0.65 (35%)	−0.44 (−24%)	0.50 (27%)	0.91

Note: The figures in brackets are the % value of the metric (either the root mean squared error RMSE, the MAE or the bias) relative to the mean AGB of trees in the dataset. NFI, National Forest Inventory.

*Derived by comparing the AGB estimates from each model (sum of AGB components) with the laboratory-measured dry weight of the trees in each dataset.

†In the case of the models developed in this study (Western Peatlands, Taiga Plains, and Boreal Plains), the AGB estimate for each tree comes from the cross-validated model that excluded the plot to which the tree belonged.

‡Ung et al. (2008).

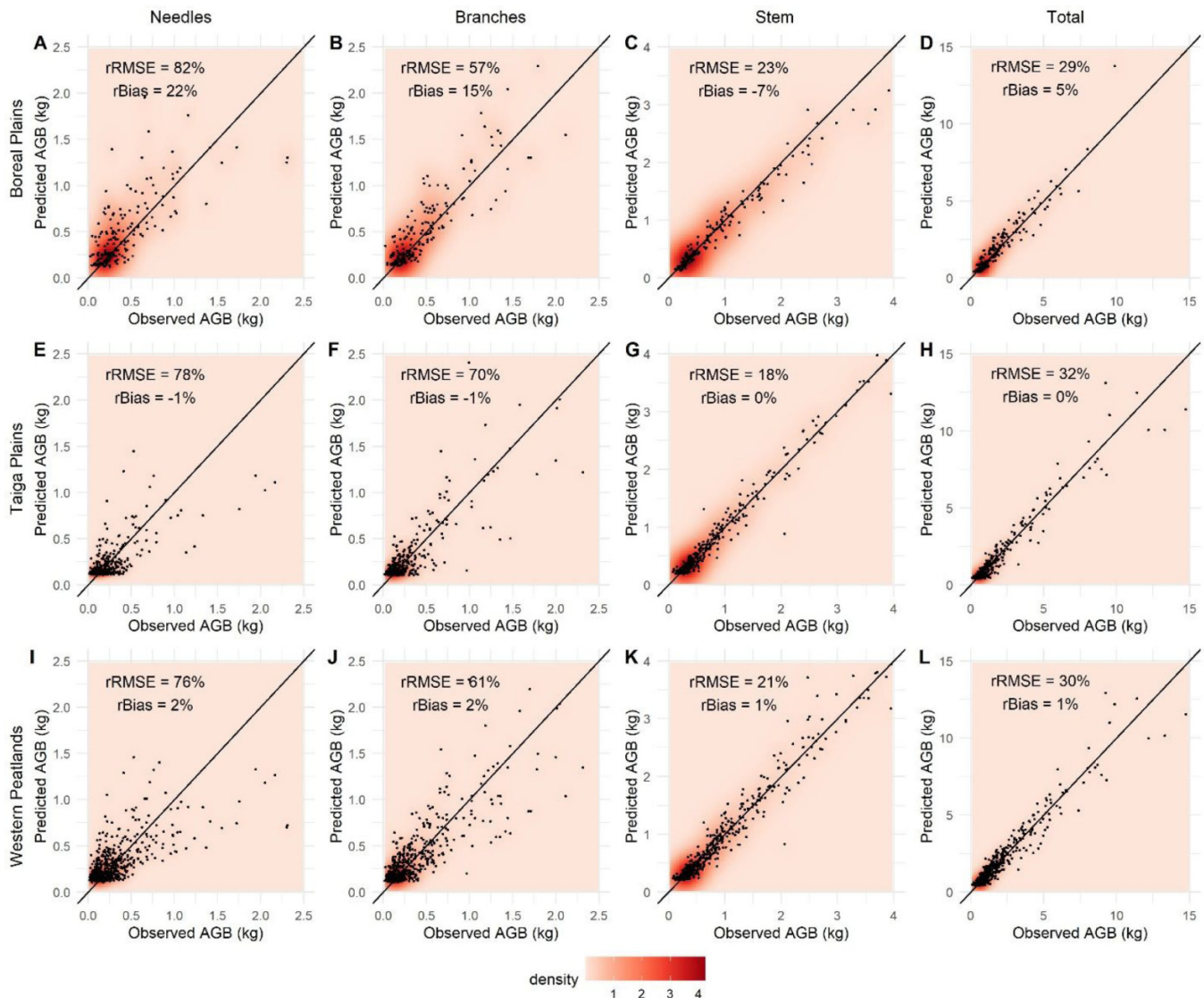
§Lambert et al. (2005a).

3.3. Objective 3: comparison with the national and NFI equations and other models

The accuracy metrics for the WP model were compared with those from the national model (Ung et al. 2008) on the 495 trees in this study, while the accuracy metrics for the TP and BP models in their respective datasets were compared with those for the national model, the customized models (Lambert et al. 2005b) used in those ecozones by the National Forest Inventory (NFI), and the WP model (Table 2). Predicted AGB was computed for each component (needles, branches, stems, and total) for each model and dataset (WP, TP, BP) based on the LOPO version of the model that excluded the plot to which each tree belonged. These predictions were then compared with the laboratory-measured AGB (Fig. 2), resulting in the relative RMSEs, relative biases, and adjusted R² values seen in Fig. 3.

The WP model performed well for all three datasets. In the TP ecozone, it had an identical RMSE, and only a slightly higher bias than the TP model. In the BP, it even outperformed the BP model. When applied on the entire 495-tree dataset, the WP model had a smaller RMSE and a lower bias than the national model (Table 2), and similar bias but much smaller RMSE than models commonly used in the USA (Table A2, upper half). When the WP model was compared to the Wagers et al. (2021) model, which in addition to height uses crown area instead of DBH as the other predictor, the latter outperformed the WP model (Table A5), corroborating these authors’ finding that crown area is a better predictor of AGB than DBH, at least for the set of 85 trees in the comparison. However, precise crown area measurements require either laser scanning or photogrammetry.

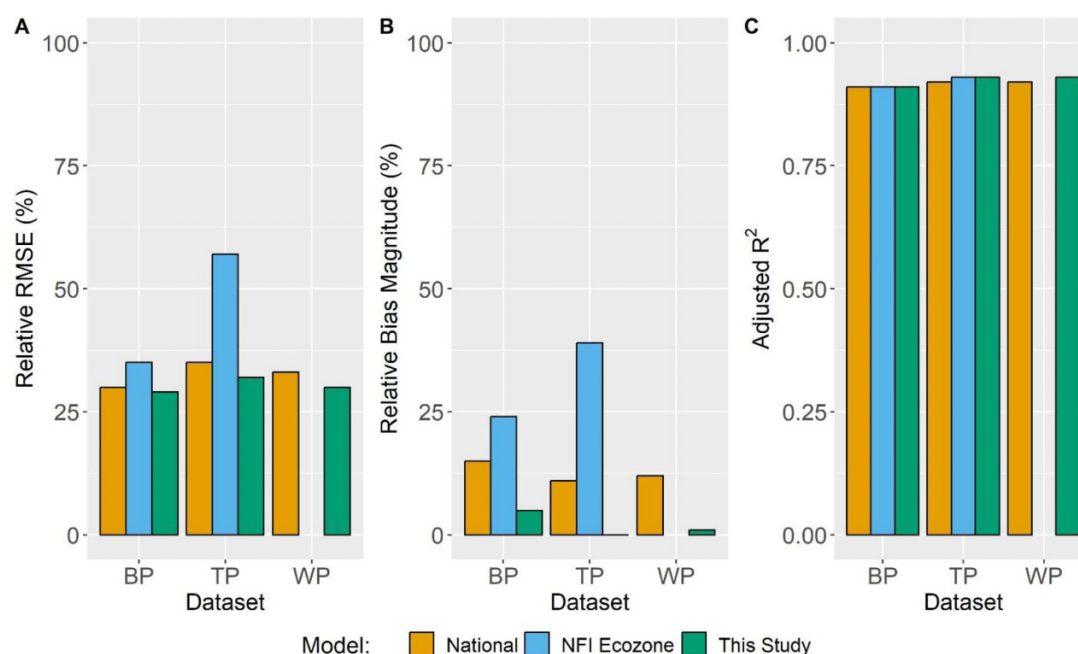
Fig. 2. Predicted versus observed scatter plots for each model (Boreal Plains, Taiga Plains, and Western Peatlands), by component. Points show the aboveground biomass (AGB) values predicted by the models for each individual tree, and areas with a high density of points are indicated with shading. Relative root mean square error (rRMSE) and relative bias (rBias) are relative to the mean laboratory-measured AGB of the trees in the particular dataset.



In the TP and BP ecozones, our models perform similarly to the national model in terms of %RMSE and adjusted R^2 , but they are considerably less biased (Fig. 3). Interestingly, the NFI-customized models, which are supposed to produce better results than the national equations in their particular geographic domain, yielded the largest RMSE values for both the TP Ecozone and the BP Ecozone. A similar pattern is seen in terms of the relative bias magnitude, where the NFI models are outperformed by the national model, and the latter by our models. The MAE results also indicate that, on average, each of the models developed in this study estimated AGB more accurately than the national model for the study sample. The full numerical results (excluding bias magnitude, which is only shown in Fig. 3) are provided in Table 2.

Regarding the influence of tree size (in terms of AGB) on the magnitude and sign of errors in both the WP and national models (Fig. 4), MAE increased with tree size in both models. However, the opposite was true in terms of relative MAE (rMAE; not shown but can be inferred from Fig. 4A): rMAE for the largest trees (Q4, where Q refers to quartile) was less (15% in both the WP and the national) than that (35% in the WP and 54% in the national) for the smallest trees (Q1). The real gain from our WP model in terms of magnitude of error was for trees below the median size (1.1 kg AGB), where MAE was much smaller than that of the national model. For trees larger than the median size, MAE in both models was about the same. However, the national model systematically underestimated AGB for all quartiles, while the WP model was almost unbiased in Q2 and Q3, with only a slight under-

Fig. 3. Bar chart of accuracy metrics for the models developed in this study, the National Forest Inventory (NFI) ecozone models (Lambert et al. 2005b), and the national model (Ung et al. 2008), in three geographic domains: the Boreal Plains (BP), the Taiga Plains (TP), and both regions combined (WP—Western Peatlands). The reported metrics are (A) relative (to mean aboveground biomass of trees in the dataset) root mean squared error (RMSE), (B) relative bias magnitude (i.e., in terms of absolute values), and (C) adjusted R^2 when the predicted AGB of the model (sum of the components) are regressed against the reference aboveground biomass values.



estimation for Q4 (-0.06 kg) and a slight overestimation for Q1 ($+0.13$ kg) (Fig. 4B).

3.4. Objective 4: upscaling to plot-level AGB estimates

Plot-level AGB, in units of $\text{Mg}\cdot\text{ha}^{-1}$, was calculated using the WP model (again using in each plot the cross-validated LOPO model that excluded that plot) and the national model following the steps outlined in Section 2.5. The WP model outperformed the national model in terms of both RMSE and bias (Table S3). The %RMSE (relative to laboratory-measured plot AGB) was 15.9% using our model, compared with 18.6% using the national model (Fig. 5). The mean plot-level bias using our model was $+0.39 \text{ Mg}\cdot\text{ha}^{-1}$ ($+1.9\%$ relative to laboratory-measured plot AGB), whereas the bias using the national model was $-2.33 \text{ Mg}\cdot\text{ha}^{-1}$ (-11.4%).

Our model leads to more accurate AGB-density estimation than the national model in 67% of the plots, which is unsurprising given the smaller bias and higher accuracy of our model at the individual tree level (Table 2). With our WP model, about half of the plots had an AGB-density error under $\pm 10\%$, which is about a third of the RMSE for the individual trees (30% of the mean tree AGB), meaning that a good portion of the individual tree over- and under-estimations cancel each other within each of those plots. There were, however, four plots ($\sim 8\%$) in which individual tree errors seemed to be added up because all of them were either over- or under-estimated. In contrast, when the national model was

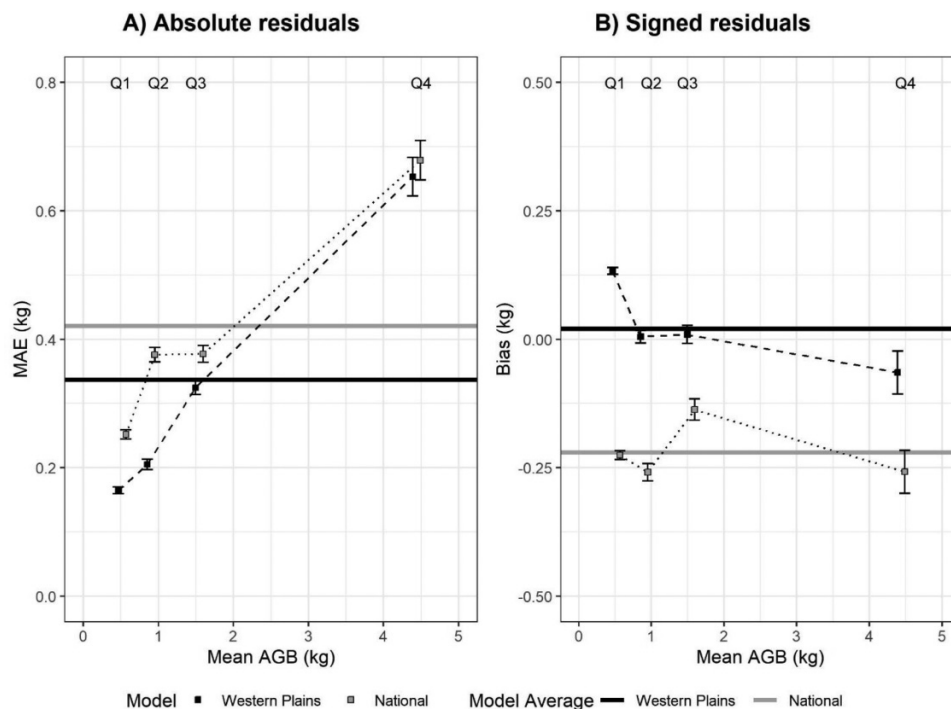
used, less than a third (27%) of plots had an error under $\pm 10\%$, and plot-level AGB was underestimated in $\sim 75\%$ of the plots.

3.5. Objective 5: effect of adding stand variables to AGB estimation models.

Two stand variables (stem density and basal area) were incorporated as additional predictors for the subset of 372 trees described in Section 2.6 (Table S4). Stem density had the most noticeable effect on the models, with the DBH-height-stem density model (eq. 6) resulting in the lowest RMSE and performing the best in terms of AIC and BIC (Table S5). It did, however, have a higher bias than both the DBH-height model (eq. 7) and the DBH-height-basal area model (eq. 8). The latter on the other hand did not outperform the DBH-height model in terms of RMSE, bias, AIC, or BIC. The DBH-height-stem density-basal area model resulted in the highest RMSE, bias, AIC, and BIC values of the four models tested. Model coefficients for these models can be found in Table S3 and results of these comparisons can be found in Table S4.

Linear regressions of ecological variables against plot-level residuals showed that stem density and basal area had significant relationships with the magnitude of plot-level errors ($\alpha = 0.05$, $p_{\text{stem density}} = 0.001$, $p_{\text{basal area}} = 0.02$), while only basal area was significantly correlated with the bias of those errors ($\alpha = 0.05$, $p_{\text{stem density}} = 0.57$, $p_{\text{basal area}} = 0.01$) (Fig. 6). Stand height and mean quadratic

Fig. 4. (A) Mean absolute error (MAE) values; and (B) bias values (mean of predicted minus observed) in each quartile of the distribution of laboratory-measured aboveground biomass (AGB) from the 495 trees in this study. Quartiles are noted at the top of each plot (Q1, Q2, etc.). Black squares denote values from the Western Peatlands model and grey boxes denote values from the national model. The horizontal lines show the mean absolute error (mean residual value) for all trees in this study (black = Western Peatlands, grey = national). National model points are offset by +0.1 kg in the x axis so that boxes do not overlap.



DBH showed no significant relationship with plot-level errors ($\alpha = 0.05$, $p_{\text{stand height, signed}} = 0.82$, $p_{\text{stand height, absolute}} = 0.30$, $p_{\text{mean quadratic DBH, signed}} = 0.19$, $p_{\text{mean quadratic DBH, absolute}} = 0.66$). Using the estimates from the DBH–height–stem density model above changed the relationship between stem density and plot-level errors (weakened with magnitude of plot-level errors (absolute), strengthened with signed plot-level errors (bias)) but did not make either relationship insignificant ($\alpha = 0.05$, $p_{\text{stem density, signed}} = 0.0005$, $p_{\text{stem density, absolute}} = 0.006$).

4. Discussion

4.1. Model development, accuracy, and transferability

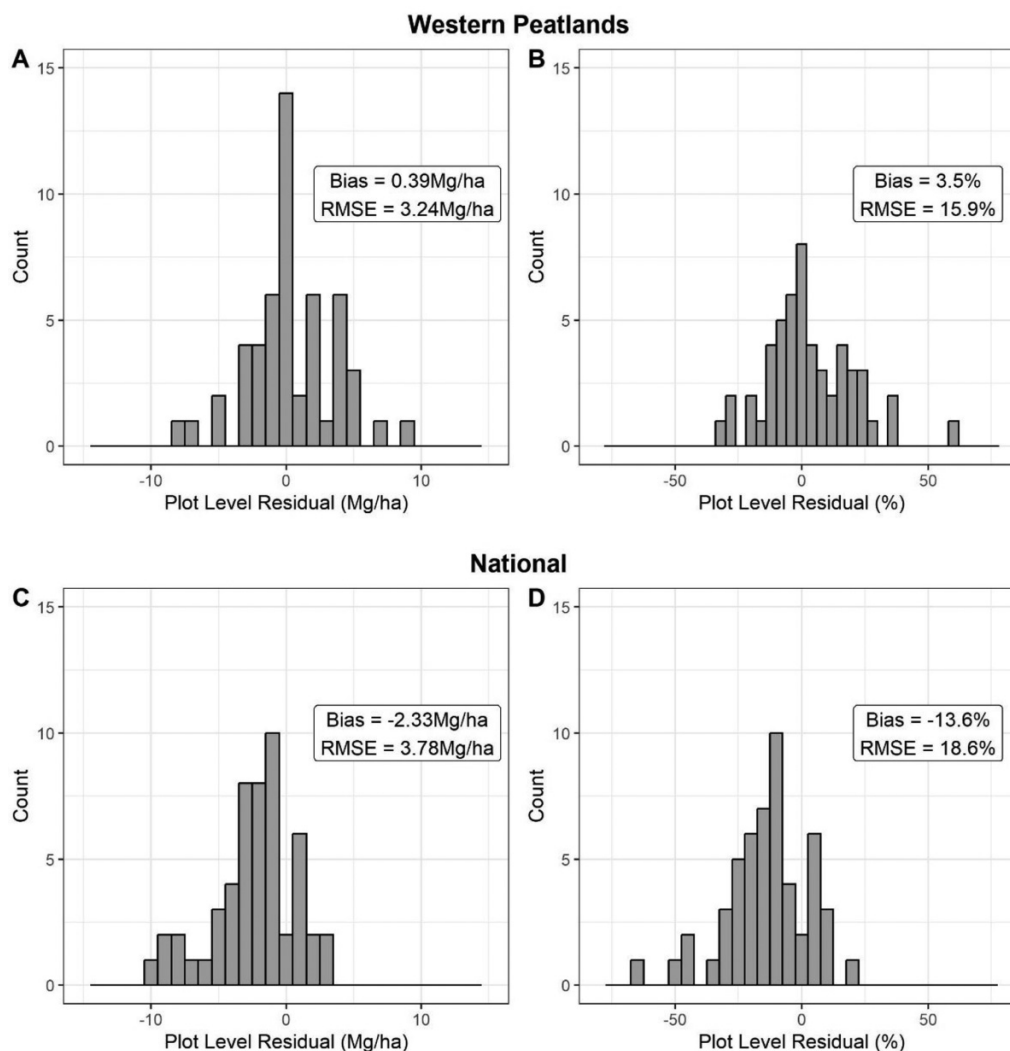
We followed the procedure and model form used by Lambert et al. (2005a) and Ung et al. (2008), which relied on SUR with DBH, and sometimes tree height, as predictor variables. Both of these tree attributes are easily measured and are commonly found in forest inventory databases (Luoma et al. 2017). Other models do exist for peatland black spruce trees, but they are for total biomass only and either rely on other predictor variables (Wagers et al. 2021), use samples that contain larger trees from non-peatland areas (Yarie et al. 2007), or were developed for use on a small local scale (Bhatti et al. 2006). Barney et al. (1978) also developed a peatland-

specific model that predicts the AGB of lowland black spruce by components, but once again they use other predictor variables in their equations (in this case basal diameter).

Unlike the national model that uses both height and DBH for all AGB components, we found that for needle and branch biomass, height was not a significant variable and even led to slightly worse predictions than those from the national equations, so we used only DBH for those two biomass components. Another difference with the national model is that the nearly 500 trees used in our study were all under 7.7 m tall (with 96% being <5 m), and they were harvested exclusively from peatlands. The addition of a positive intercept, which is absent in the national model, improved performance slightly for the smaller trees. That intercept corresponds to the expected stem, branch, or needle biomass of trees of 1.3 m height. One potential limitation in our study is that while collecting data, the plots were selected to be close to highways, which was unavoidable due to logistic constraints. As such, there is the potential for researcher bias and edge effects that may alter the performance of our models in plots away from road corridors, something future studies would need to rule out.

While DBH and height are commonly used as predictors in many other existing AGB models (Snorrason et al. 2006; Xiang et al. 2016; He et al. 2018; Latifah et al. 2021), they are not necessarily the best. Wagers et al. (2021) suggested that crown area and height may result in more accurate

Fig. 5. Histograms of residuals of the plot-level aboveground biomass (AGB) estimates from the Western Peatlands (WP) model and the national model (Ung et al. 2008) for the set of 52 plots that had stem density data in this study. Each graph includes a box with the bias (predicted—observed) and the root mean squared error (RMSE). (A) The WP model in terms of $\text{Mg}\cdot\text{ha}^{-1}$; (B) the WP model in terms of percentage of the average laboratory-measured plot AGB; (C) the national model in terms of $\text{Mg}\cdot\text{ha}^{-1}$; (D) the national model in terms of percentage of the mean laboratory-measured plot AGB.



biomass estimates, because they are less correlated among themselves than DBH and height. The comparison we did using the 85 trees common to this study and theirs (Section A5 in the Appendix), which had precise crown area measurements derived from Terrestrial Laser Scanning (TLS), corroborates what they found for the national model: the crown area and height model (total tree AGB) outperforms the WP model in terms of relative RMSE (22% versus 27%) and relative bias (-1% versus $+7\%$) (Table A5).

Ecosystem-specific models have been studied in other environments and have shown to improve on general models while still providing high transferability to other regions, provided that they are still used in the same kind of ecosystem. Nyamukuru et al. (2023) showed that this was the case for small trees in the African savannah and Abich et al. (2019) did the same for four species of trees in dry deciduous woodlands. A similar study also showed

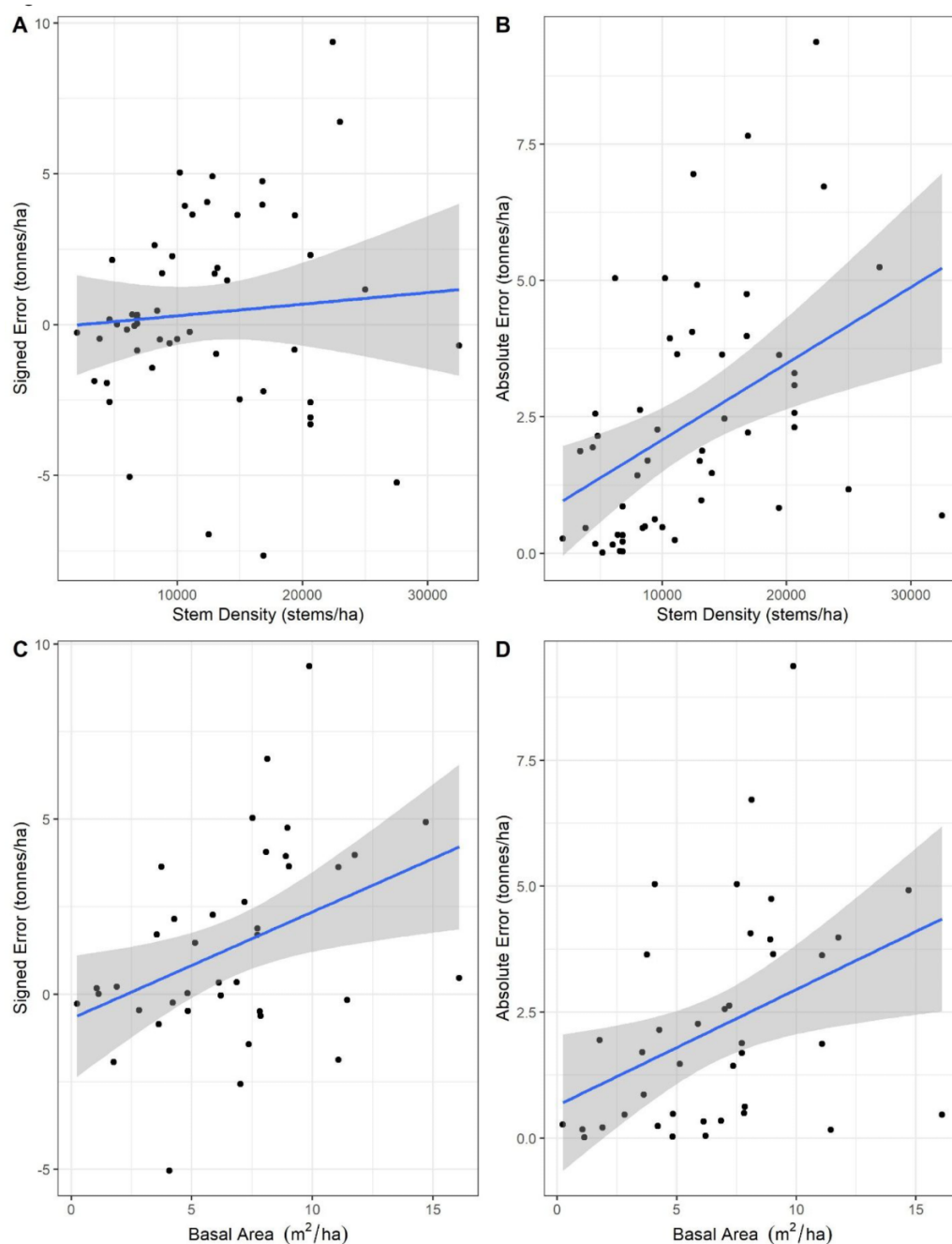
that developing ecosystem-specific models could help improve AGB estimates for shrubs in desert grasslands as well (Yang et al. 2017).

The LOPO cross-validation technique used here also provides a proxy for assessing the transferability of our models (Wenger and Olden 2012). It has also been used in other AGB studies effectively (Wagers et al. 2021; Campbell et al. 2023). Because each iteration involves fitting a model using all but 1 plot and then testing that fitted model on the removed plot, it is effectively able to test how the model would perform on a new plot that has not been seen before.

4.2. Comparison with customized models used by NFI

Another noteworthy observation from this study is the relatively poor performance of the customized (by ecozone) NFI models when compared with the national model (Table 2,

Fig. 6. Ecological variables (top: stem density; bottom: basal area) and their effect on signed (A and C) and absolute (B and D) residuals at the plot level. Blue lines indicate the regression line and shaded area indicates the 95% confidence intervals of the regression.



TP and BP datasets), which is consistent with [Case and Hall \(2008\)](#), who also found no improvement for regional equations for black spruce over the national model (Supplementary Materials, Figure S4). One possible explanation is that the national equations for black spruce were fitted using 1591 trees ([Ung et al. 2008](#)), whereas the customized versions used 32 trees for the TP Ecozone and 76 for the BP (figures inferred from the ENFOR dataset, available from <https://open.canada.ca/en/open-data>, and confirmed by the NFI of-

fice). While these subsets contain in theory the full range of tree sizes in that ecozone, they have far fewer small trees than the full national dataset to fit the lower range AGB of trees like those seen in peatlands. Precisely because our models are specific to peatland black spruce, which are often stunted and smaller than upland black spruce ([Smith et al. 2007](#)), they show hardly any bias compared with the NFI or the national models, leading to more accurate results when upscaled to the plot level.

4.3. Estimation of AGB density

When the WP model was used to estimate plot-level biomass (i.e., AGB density in $\text{Mg}\cdot\text{ha}^{-1}$), it provided better estimates and was less biased than the national model. We did not attempt an ecozone-specific comparison with the NFI-customized models, but given their higher bias, the expectation is that they will perform worse than the national model. The fact that the relative RMSE and bias of AGB estimates at the plot level were smaller than at the individual tree level (Table 2, WP dataset) clearly indicates that over- and under-estimations at the individual tree level cancelled each other out to some extent for both our model and the national model. Depending on the plot, this can range from full cancellation to hardly any cancellation (Fig. 5). This stresses the importance of having an unbiased and accurate model when estimating AGB density. Based on this evidence and the other comparisons in the Appendix (A2 and A3), we conclude that our models provide a better alternative than the national models of Ung et al. (2008), the customized models of Lambert et al. (2005b), or the American models of Barney et al. (1978) and Yarie et al. (2007) for creating high-quality AGB reference data in peatlands of Western North America populated with small black spruce.

4.4. Recommendations and applicability to other ecozones

The similarity between black spruce trees from different regions is reflected in the accuracy metrics obtained when the TP and BP models were applied to each other's ecozone. When we compared those metrics with those from the LOPO cross-validation, we observed relative RMSE differences of 3% or less, and relative bias magnitude differences of 10% or less, indicating again that the black spruce trees found in the TP Ecozone and the BP Ecozone are overall allometrically similar. Furthermore, the WP model performs equally well or better than the TP and BP models in their respective ecozones. Hence, we recommend the WP model for estimating AGB by components of small (1.3 to 5 m tall) black spruce in peatlands of the TP Ecozone and the BP Ecozone or when analyzing a wide-ranging sample consisting of trees from both ecozones. The results from the Alaskan dataset (Table A2, lower half) suggest that the WP model will also perform the best in black spruce peatlands from other ecozones, but this needs to be corroborated by further studies.

Other studies have shown that competition between trees can have some effect on the allometric relationships between tree attributes (Forrester et al. 2017b; del Río et al. 2019). These effects were observed in this study at two levels. At the individual tree level, the inclusion of stem density as an additional predictor led to slightly better accuracy (reduction of 2 points in %RMSE). However, it slightly worsened the overestimation bias by 1 point. Since bias is more important when upscaling to plot level, we do not recommend the use of this model and instead the WP model using just DBH and height seems preferable.

At the plot level, stem density and basal area showed significant positive correlations with plot-level errors (Fig. 6). While stem density only affected the error magnitude, basal area

was significantly correlated with both the magnitude and direction (overestimation) of the error. Basal area can be easily estimated in the field with a prism sweep. Hence, the regressions on basal area (Figs. 6C and 6D) can be used to quickly estimate the magnitude of error to be expected. For example, when our WP model is applied to peatland plots that have a basal area under $5 \text{ m}^2\cdot\text{ha}^{-1}$, the expected absolute error in AGB density should be less than $1.8 \text{ tonnes}\cdot\text{ha}^{-1}$, whereas if the plot has a basal area under $10 \text{ m}^2\cdot\text{ha}^{-1}$, the expected absolute error in AGB density should be less than $3.0 \text{ tonnes}\cdot\text{ha}^{-1}$. When we developed models that contained stem density and basal area as predictors, only stem density improved the AGB estimations. At the plot level, however, there was still a significant relationship between plot-level residuals and stem density. Based on this and the higher bias that was seen at the individual tree level, we recommend again the more parsimonious WP model as opposed to the model containing stem density.

An interesting question we could not tackle is whether a warming climate could alter the allometric relationships our equations describe. The distribution pattern of our plots, where any given plot is often just a few kilometers away from the nearest plot, does not provide enough data points with different climate conditions to assess this. Notwithstanding, it is unlikely that we would find any effect if we had more distant plots. Evergreen trees such as spruce are less sensitive to the variation of climatic conditions in terms of AGB accumulation than deciduous trees (Wilson and Elling 2004). In fact, Xu et al. (2021) incorporated climatic variables to conventional allometric models of black spruce fitted with 783 trees from 54 sites across Canada and found no improvement in biomass estimation (however, they did find improvements for tamarack and trembling aspen, which are deciduous).

5. Conclusions

In this study, we used a new dataset of nearly 500 trees to develop allometric models of individual tree AGB by components (stem, branches, and needles) for small (<5 m) black spruce specific to one of the most common ecosystems of the Canadian boreal forest: treed peatlands. Our models outperformed the national and ecozone-specific models currently being used by NFI, both in our tree dataset and in other datasets not necessarily from peatlands. Adding stem density as a predictor slightly decreased RMSE but increased bias. When our WP model was used to upscale to plot-level estimates (in units of $\text{Mg}\cdot\text{ha}^{-1}$), the WP model outperformed the national model again, largely due to their nearly unbiased predictions. Hence our models are currently the best option to create high-quality reference AGB data using ground-measured predictor variables for peatlands dominated by small black spruce trees. When we applied our model to black spruce in interior Alaska (Appendix A2), our model yielded less error than previously existing models for Alaskan black spruce. Application to black spruce peatlands in Eastern Canada should lead to similar results, but this would need to be corroborated by further studies.

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Data availability

Data generated or analyzed during this study are available in the zenodo repository, [10.5281/zenodo.7362485, zenodo.org/record/7362485] (Voicu and Rea 2023). Data on plot size in this study are available from Mihai Voicu (mihai.voicu@canada.ca) upon reasonable request.

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Competing interests

The authors declare there are no competing interests.

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Supplementary material

Supplementary data are available with the article at <https://doi.org/10.1139/cjfr-2023-0031>.

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Appendix: Comparing the Western Peatlands model with other existing models

A1. Summary information on the models compared

In addition to the comparisons made between the models developed in this study and the national and regional mod-

Table A1. A summary of the existing allometric models that were compared with the Western Peatlands model developed in this study.

Model	No. of trees used to fit the model	Location of trees used to fit the model, and model characteristics
Barney et al. (1978)	18	Alaska, USA (upland and lowland sites). Only the lowland model was compared. Model form: $\ln(\text{AGB}_i) = a + b \ln(\text{BD})$, where AGB is the aboveground biomass of component i , BD is the basal diameter of the tree, and a and b are model coefficients.
Yarie et al. (2007)	126 (40)*	Interior Alaska. Out of the several models available for black spruce in their report, only the one yielding the best results in the comparison is shown. Model form: $\text{AGB}_i = a + b \cdot \text{BD}^2$, where AGB is the aboveground biomass of component i , BD is the basal diameter of the tree, and a and b are model coefficients.
Ung et al. (2008)	1591 (257) [†]	Across Canada (mostly from managed forests, and many not from peatlands). Here we only report the comparison that used the ENFOR calibration subset with black spruce trees < 5 m tall (for the comparison with our tree dataset, see Section 3.3).
Wagers et al. (2021)	86	Northwest Territories, Canada (peatland sites). These 86 trees are from the 495 in this study that had precise crown area measurements. The model has same form as Ung et al. (2008) but it is for total tree biomass only and uses crown area instead of DBH.
Fradette et al. (2021)	96 (60) [‡]	Quebec, Canada (open woodland site). Model form: $\text{AGB}_i = a \cdot \text{DBH}^b \cdot H^c$, where AGB is the aboveground biomass of component i , DBH is the diameter at breast height, H is the height of the tree, and a , b , and c are model coefficients.

*Note: All Models were compared using the dataset in this study. In addition, some models for which we could obtain their calibration tree data were subject to a second comparison using the subset of their trees that were shorter than 5m (figure in bracket in second column).

[†]Dataset available from Ung et al. 2017 (doi: 10.23687/FBAD665E-8AC9-4635-9F84-E4FD53A6253C).

*Dataset available from Mack 2014 (doi: 10.6073/PASTA/AECC1EE8FEF1D76C8DCB5D24E492AFE7).

[‡]Dataset obtained from authors.

els (Lambert et al. 2005b; Ung et al. 2008), we also compared the Western Peatland (WP) model's performance in terms of total tree biomass against other existing models used in the United States and Canada to estimate the aboveground biomass (AGB) of black spruce trees. A summary of the models used for this comparison can be found in Table A1. The details of each comparison are noted in the subheadings below.

A2. Comparisons with American models

Two commonly used American models were compared with the WP model presented in this study. AGB estimates for both our 495-tree dataset and a public dataset of small black spruce in interior Alaska (mean height 3.0 m; mean DBH 4.4 cm, mean AGB 4.1 kg) (Mack, 2014) were obtained using the Barney model (Barney et al. 1978), the Yarie model (Yarie et al. 2007), and the WP model. RMSE, bias, and the adjusted R^2 of predicted values versus observed values were computed and compared for each dataset (Table A2).

When applied on the entire 495-tree dataset, the WP model had the lowest RMSE of the four models tested (WP residuals derived from LOPO cross-validation). It also outperformed the national and Yarie models in terms of bias. The Barney model resulted in the same magnitude of bias as the WP model, but with a much higher RMSE. When tested on black spruce trees <5 m tall in similar ecosystems in Alaska, our model again outperformed the local Yarie model in terms of RMSE. It also predicted much better than the Barney model and the national model, both in terms of bias and RMSE.

After scaling the AGB estimates up to the plot level (using the 39 plots from this study that had stem density data), the relative RMSE (relative to laboratory-measured plot AGB) was 21.0% using the Yarie model, and 23.9% using the Barney model, both higher than the WP model (15.9%) or the

Table A2. Accuracy metrics (RMSE, bias, and adjusted R^2) for aboveground biomass (AGB) estimates from the Western Peatlands model developed in this study, the national model (Ung et al. 2008), and the models presented in Yarie et al. (2007) and Barney et al. (1978) for the 495-tree dataset in this study and for a dataset in interior Alaska (Mack 2014).

Model	RMSE (kg)	Bias (kg)	Adjusted R^2
<i>Dataset: 495-tree dataset from this study</i>			
Western Peatlands	0.53 (30%)	0.02 (+1%)	0.93
National	0.60 (33%)	-0.22 (-12%)	0.92
Yarie et al.	0.67 (37%)	0.06 (+3%)	0.89
Barney et al.	0.74 (41%)	-0.02 (-1%)	0.89
<i>Dataset: 40-tree interior Alaska dataset (Mack, 2014)</i>			
Western Peatlands	1.86 (41%)	0.82 (+18%)	0.92
National	3.69 (81%)	2.14 (+47%)	0.90
Yarie et al.	2.71 (60%)	0.85 (+19%)	0.88
Barney et al.	5.93 (131%)	2.85 (+63%)	0.87

Note: Percent values relative to the mean AGB of the dataset are noted in brackets.

national model (18.6%) (Fig. 5). The WP model also had the lowest plot-level bias (+0.39 Mg·ha⁻¹ or +1.9% relative to laboratory-measured plot AGB), whereas the bias using the national model was -2.33 Mg·ha⁻¹ (-11.4%), +0.48 Mg·ha⁻¹ (+2.3%), and -0.89 Mg·ha⁻¹ (-4.4%), respectively. Even though the Barney and Yarie models were less biased than the national model for small black spruce trees in peatlands, they predicted fewer plots with an error under $\pm 10\%$ than the WP model (44% and 32% of the plots, respectively, whereas for the WP model is 50%). Thus, the WP model should be the model of choice for practitioners estimating biomass in

Table A3. Accuracy metrics for aboveground biomass (AGB) estimates made using the Western Peatlands model developed in this study and the national model (Ung et al. 2008) for the 257 black spruce trees from the ENFOR dataset with heights between 1.3 and 5 m.

Model	RMSE (kg)	Bias (kg)	Adjusted R ²
Western Peatlands	1.51(40%)	0.01 (1%)	0.76
National	1.49(40%)	0.02 (2%)	0.79

Note: Values of root mean squared error (RMSE) and bias are given in kg and in brackets as a percentage of the average AGB of trees in the sample.

Table A4. Comparison of the Western Peatlands model, the national model, and a local model given by Fradette et al. (2021) in a subset of 60 black spruce trees between 1.3 and 5 m in height from the open woodlands dataset used by those authors to develop their model in Quebec, Canada (dataset obtained from authors).

Model	RMSE (kg)	MAE (kg)	Bias (kg)
Western Peatlands	4.86 (86%)	3.07 (54%)	−2.95 (−52%)
National	4.87 (86%)	3.24 (57%)	−3.05 (−54%)
Fradette (2021)	3.39 (60%)	2.28 (40%)	0.49 (9%)

boreal peatland environments, either in Western Canada or Alaska.

A3. Comparisons using the ENFOR dataset

The national model (Ung et al. 2008) was developed using data from the Energy From the Forest (ENFOR) program. Because the trees in the ENFOR dataset were harvested in different environments from all over Canada including upland forest, we were able to test the applicability of our model to small black spruce not in peatlands. For this, we used the 257 black spruce trees from the ENFOR dataset that were between 1.3 and 5 m tall (mean height 3.4 m; mean DBH 3.8 cm, mean AGB 3.8 kg), the application range we recommend for our model. RMSE, bias, and the adjusted R² of observed values versus predicted values were calculated to compare the WP and national models (Table A3).

Our model performed similarly to the national model even though these trees were part of the dataset used to fit the latter and even had slightly less biased estimates. This suggests the WP model may be preferable to the national model on any small black spruce trees regardless of ecosystem, but this does not imply that the WP model is preferable to a local model developed for a non-peatland ecosystem, as we show next.

A4. Comparisons in an open woodland ecosystem

In order to test how the WP model performs in a particular non-peatland environment, we applied it to a dataset consisting of 60 trees destructively sampled in a 300 ha study area of mesic, open woodland in Quebec, Canada, which was the subset of Fradette et al. (2021) black spruce trees between 1.3

Table A5. Accuracy metrics for total tree aboveground biomass (AGB) estimates from the Western Peatlands model developed in this study and the Wagers et al (2021) model that uses crown area instead of DBH as one of the predictors (the other is height), based on 85 trees common to both studies.

Model	RMSE (kg)	Bias (kg)	Adjusted R ²
Western Peatlands	0.46 (27%)	+0.12 (+7%)	0.94
Wagers	0.37 (22%)	−0.01 (−1%)	0.96

Note: Values of root mean squared error (RMSE) and bias are given in kg, and in brackets as a percentage of the average AGB of trees in the sample.

and 5 m tall (mean height 2.8 m, mean DBH 2.8 cm, and mean AGB 5.7 kg). Estimates were made using the WP model, the national model, and the local model given by those authors. RMSE and bias were used as accuracy metrics to compare the models. Differences in MAE of the models were tested for significance using a paired one-tailed Wilcoxon signed-rank test.

In this comparison, the WP model performed similarly to the national model, but had a slightly lower MAE (Table A4). However, the local model that Fradette et al. (2021) fitted using 96 trees, including those 60, performed the best for all three metrics.

The low stem density found in open woodland plots favours less competition for nutrients and sunlight amongst the trees. Many studies have shown that trees growing in these conditions have differing allometries than those growing in denser stands (Mäkelä and Valentine 2006;Forrester et al. 2017a;Forrester et al. 2017b; del Río et al. 2019). The large underestimation that the WP and the national models incurred suggests that the growth of black spruce in open woodlands differs from the general growth trends in other ecosystems. This is consistent with the metanalysis of Grigal and Kernik (1984) on the generality of black spruce biomass equations, who found that the Rencz and Auclair (1980) study in an open spruce-lichen woodland in Quebec had the most different relationship in both total and foliar biomass among the dozen studies they considered that included both upland and peatland.

A5. Comparison with a model that uses crown area instead of DBH as predictor

To ascertain whether the superiority of crown area as predictor that Wagers et al. (2021) found for total tree biomass of small black spruce still holds against our enhanced DBH-based model, we compared the WP model with their model in an 85-tree subset of our 495-tree dataset (mean height 2.7 m, mean DBH 2.7 cm, and mean AGB 1.8 kg). These trees were scanned using a terrestrial laser scanner (TLS) and the resultant point clouds were used to estimate the crown area of each individual tree using the raster method outlined in Wagers et al. (2021). The resulting RMSEs and relative biases for both models are consistent with the conclusion of that paper that crown area and height are better AGB predictors than DBH and height (Table A5). Thus, practitioners only interested in total tree biomass of small black spruce rather than biomass by components can accurately estimate the AGB of individual trees using crown area and height de-

rived from high-density point clouds from drone photogrammetry or LiDAR, thereby bypassing the need for ground measurements.

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